

Date: 20/12/24

Lab Practical #02:

Perform NMAP tool for scanning system vulnerability.

1. nmap -p :

- port scanning port specific port

```
(kali@kali)-[~]  
$ nmap -p 23 172.16.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:26 EST  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.04 seconds
```

2. nmap -p :

- port scanning port specific port range

```
(kali@kali)-[~]  
$ nmap -p 23-100 172.16.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:32 EST  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.05 seconds
```

3. nmap -pU:110, T:23-25, 443 :

- U-UDP, T-TCP different port type scan

```
(kali@kali)-[~]  
$ nmap -pU:110,T:23-25,443 172.16.1.1  
  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:44 EST  
WARNING: Your ports include "U:" but you haven't specified UDP scan with -sU.  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds
```

Date: 20/12/24

4. nmap -p- :

- port scan for all ports

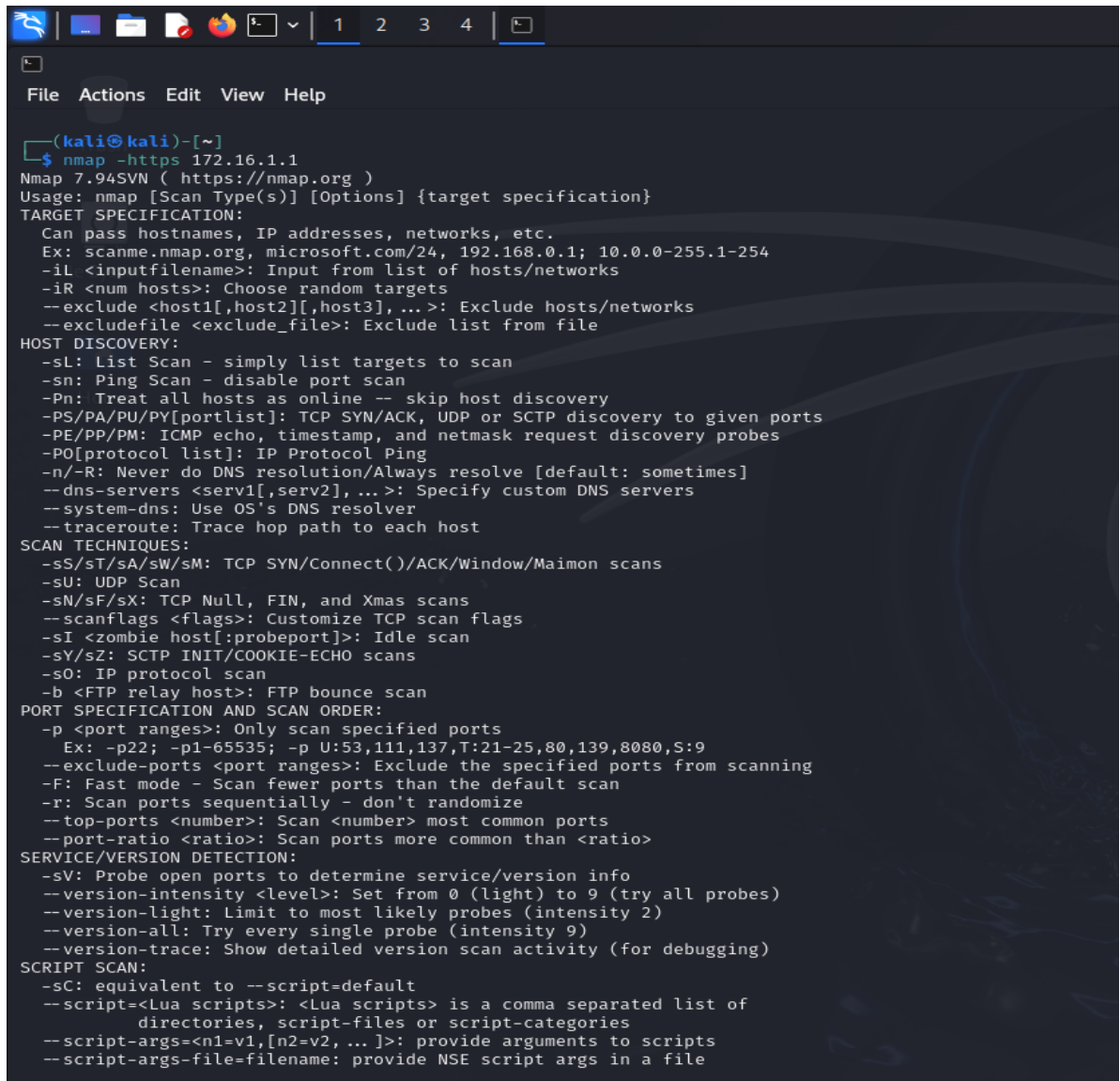
```
(kali@kali)-[~]
$ nmap -p- 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:48 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.04 seconds

(kali@kali)-[~]
$
```

5. nmap -smtp, https :

- port scan for specified protocols

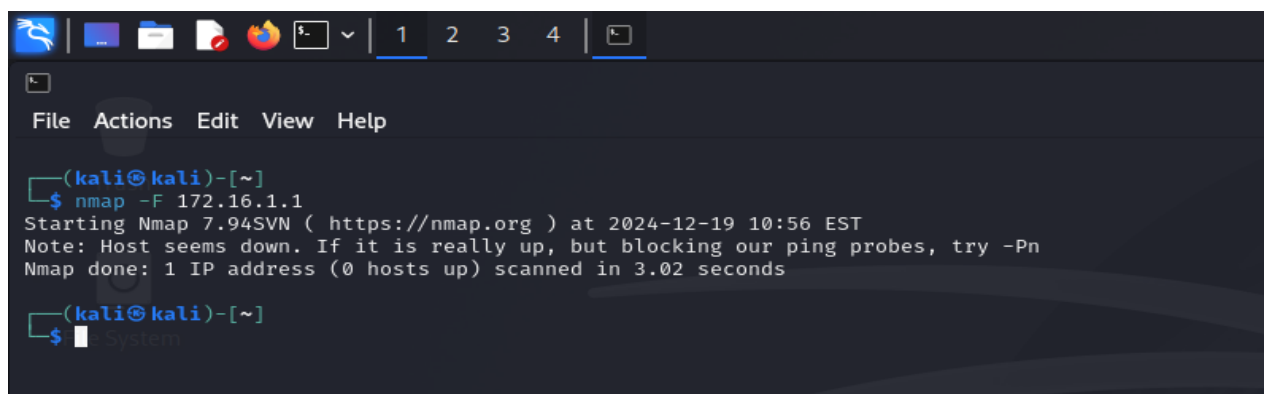
```
(kali@kali)-[~]
$ nmap -smtp 172.16.1.1
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sw/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
    directories, script-files or script-categories
  --script-args=<n1=v1[,n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
```



```
(kali㉿kali)-[~]
$ nmap -https 172.16.1.1
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script-default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
    directories, script-files or script-categories
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
```

6. nmap -F :

- fast port scan for speed up



```
(kali㉿kali)-[~]
$ nmap -F 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:56 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.02 seconds

(kali㉿kali)-[~]
$
```

Date: 20/12/24

7. nmap -p "*" :

- port scanning using name

```
(kali㉿kali)-[~]
$ nmap -p "*" 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 10:58 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds

(kali㉿kali)-[~]
$ █
Home
```

8. nmap -r :

- sequential port scan

```
(kali㉿kali)-[~]
$ nmap -r 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:01 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds

(kali㉿kali)-[~]
$ █
```

9. nmap -sS :

- TCP SYN port scan

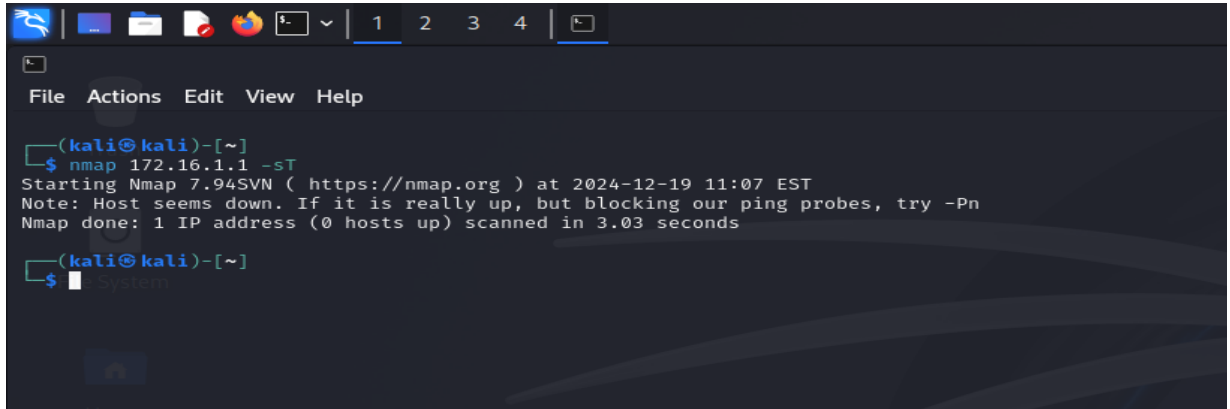
```
(kali㉿kali)-[~]
$ nmap -sS 172.16.1.1
You requested a scan type which requires root privileges.
QUITTING!

(kali㉿kali)-[~]
$ █
```

Date: 20/12/24

10. nmap -sT :

- TCP connect port scan

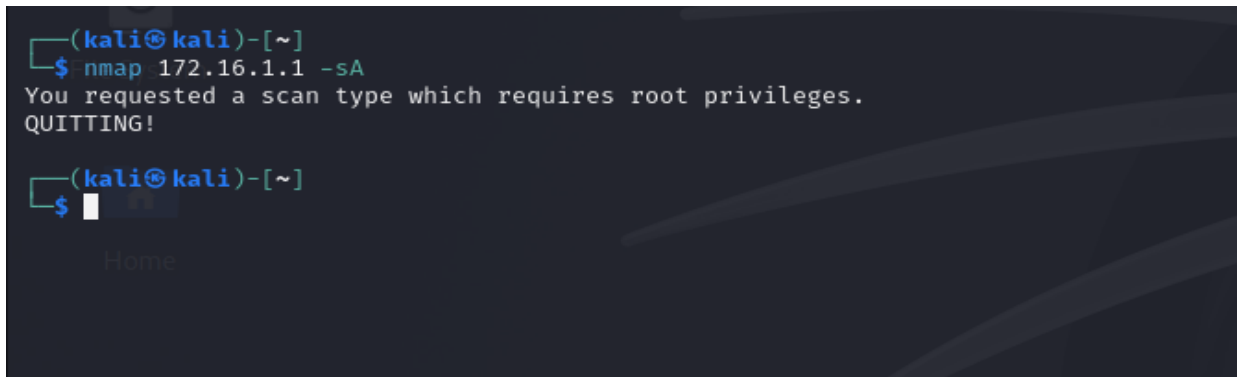


```
(kali㉿kali)-[~]
$ nmap 172.16.1.1 -sT
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:07 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds

(kali㉿kali)-[~]
$
```

11. nmap -sA :

- TCP ACK port scan

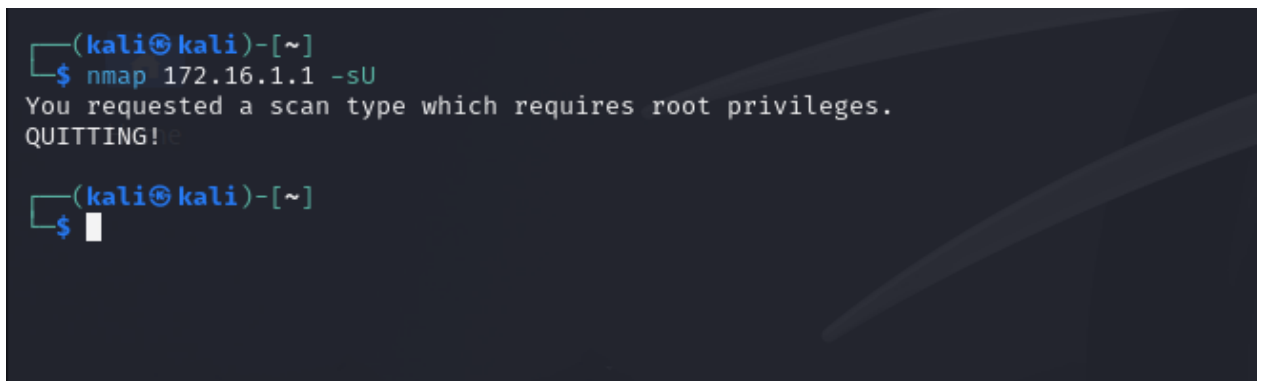


```
(kali㉿kali)-[~]
$ nmap 172.16.1.1 -sA
You requested a scan type which requires root privileges.
QUITTING!

(kali㉿kali)-[~]
$
```

12. nmap -sU :

- UDP port scan



```
(kali㉿kali)-[~]
$ nmap 172.16.1.1 -sU
You requested a scan type which requires root privileges.
QUITTING!

(kali㉿kali)-[~]
$
```

Date: 20/12/24

13. nmap -sF :

- TCP FIN scan

```
(kali㉿kali)-[~]  
$ nmap -sF 172.16.1.1  
You requested a scan type which requires root privileges.  
QUITTING!  
  
(kali㉿kali)-[~]  
$
```

14. nmap -sX :

- XMAS scan

```
(kali㉿kali)-[~]  
$ nmap -sX 172.16.1.1  
You requested a scan type which requires root privileges.  
QUITTING!  
  
(kali㉿kali)-[~]  
$
```

15. nmap -sP :

- ping scan

```
(kali㉿kali)-[~]  
$ nmap -sP 172.16.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:16 EST  
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn  
Nmap done: 1 IP address (0 hosts up) scanned in 3.01 seconds  
  
(kali㉿kali)-[~]  
$
```

Date: 20/12/24

16. nmap -sU :

- UDP scan

```
(kali㉿kali)-[~]  
$ nmap -sU 172.16.1.1  
You requested a scan type which requires root privileges.  
QUITTING!  
  
(kali㉿kali)-[~]  
$
```

17. nmap -sA :

- TCP ACK scan

```
(kali㉿kali)-[~]  
$ nmap -sA 172.16.1.1  
You requested a scan type which requires root privileges.  
QUITTING!  
  
(kali㉿kali)-[~]  
$
```

18. nmap -sL :

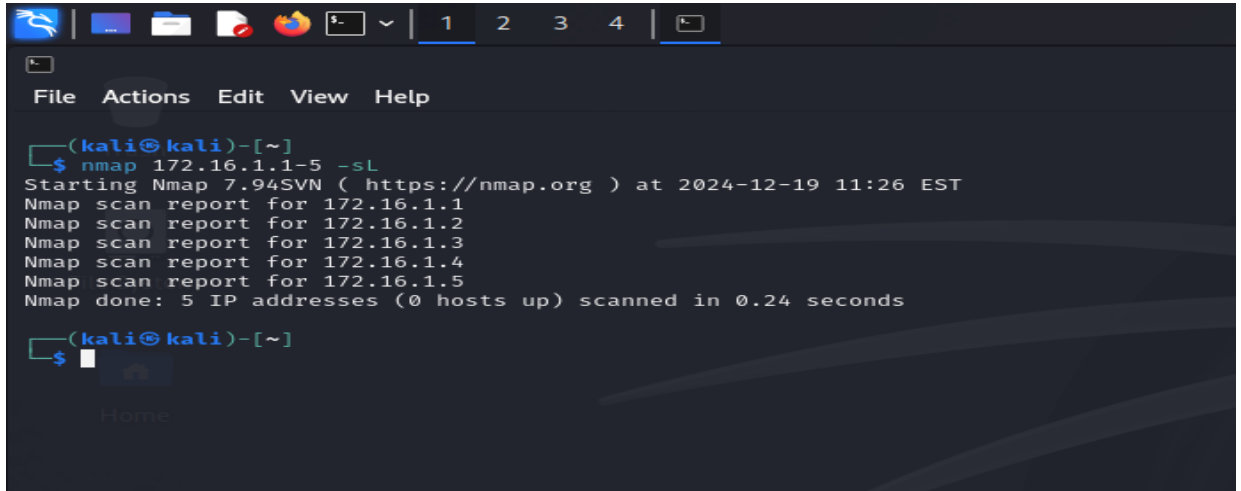
- list scan

```
(kali㉿kali)-[~]  
$ nmap -sL 172.16.1.1  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:24 EST  
Nmap scan report for 172.16.1.1  
Nmap done: 1 IP address (0 hosts up) scanned in 0.08 seconds  
  
(kali㉿kali)-[~]  
$
```

Date: 20/12/24

19. nmap -sL :

- list 172.16.1.1 without scanning

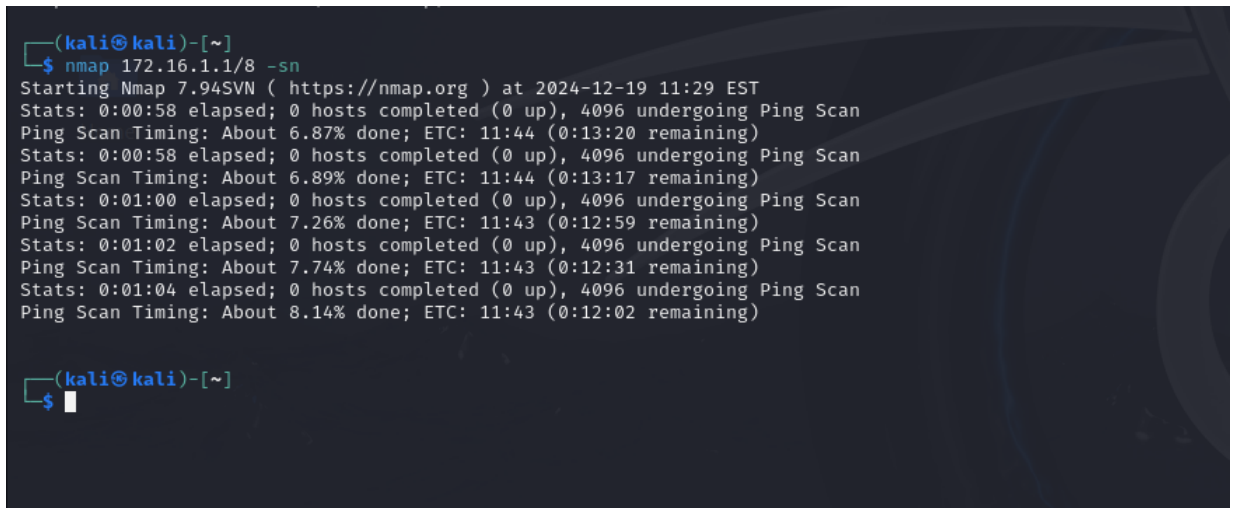


```
(kali㉿kali)-[~]
$ nmap 172.16.1.1-5 -sL
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:26 EST
Nmap scan report for 172.16.1.1
Nmap scan report for 172.16.1.2
Nmap scan report for 172.16.1.3
Nmap scan report for 172.16.1.4
Nmap scan report for 172.16.1.5
Nmap done: 5 IP addresses (0 hosts up) scanned in 0.24 seconds

(kali㉿kali)-[~]
$
```

20. nmap -sn :

- disable port scanning



```
(kali㉿kali)-[~]
$ nmap 172.16.1.1/8 -sn
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:29 EST
Stats: 0:00:58 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 6.87% done; ETC: 11:44 (0:13:20 remaining)
Stats: 0:00:58 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 6.89% done; ETC: 11:44 (0:13:17 remaining)
Stats: 0:01:00 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 7.26% done; ETC: 11:43 (0:12:59 remaining)
Stats: 0:01:02 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 7.74% done; ETC: 11:43 (0:12:31 remaining)
Stats: 0:01:04 elapsed; 0 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 8.14% done; ETC: 11:43 (0:12:02 remaining)

(kali㉿kali)-[~]
$
```


Date: 20/12/24

21. nmap -Pn :

- port scans only and no host discovery

```
(kali@kali)-[~]
$ nmap 172.16.1.1-8 -Pn
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-19 11:32 EST
Stats: 0:00:02 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.06% done
Stats: 0:00:06 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.31% done
Stats: 0:00:07 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.38% done
Stats: 0:00:08 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.44% done
Stats: 0:00:09 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.50% done
Stats: 0:00:11 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.62% done
Stats: 0:00:13 elapsed; 0 hosts completed (8 up), 8 undergoing Connect Scan
Connect Scan Timing: About 0.75% done
```

22. nmap -PS22-25, 80 :

- TCP SYN discovery on specified port

```
(kali@kali)-[~]
$ nmap 172.16.1.185 -PS22-25,80
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:10 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.07 seconds

(kali@kali)-[~]
$
```

23. nmap -PA22-25, 80 :

- TCP ACK discovery on specified port

```
(kali@kali)-[~]
$ nmap 172.16.1.185 -PA22-25,80
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:13 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.03 seconds

(kali@kali)-[~]
$
```

Date: 20/12/24

24. nmap -PR :

- ARP discovery within local network

```
(kali㉿kali)-[~]
$ nmap 172.16.1.1-1/8 -PR
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:17 EST
Stats: 0:00:04 elapsed; 40960 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 31.88% done; ETC: 08:17 (0:00:00 remaining)
Stats: 0:00:05 elapsed; 49152 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 70.92% done; ETC: 08:17 (0:00:00 remaining)
Stats: 0:00:05 elapsed; 57344 hosts completed (0 up), 4096 undergoing Ping Scan
Ping Scan Timing: About 79.37% done; ETC: 08:17 (0:00:00 remaining)

(kali㉿kali)-[~]
$
```

25. nmap -n :

- no DNS resolution

```
(kali㉿kali)-[~]
$ nmap 172.16.1.1 -n
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:21 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.03 seconds

(kali㉿kali)-[~]
$
```

26. nmap --script= test script :

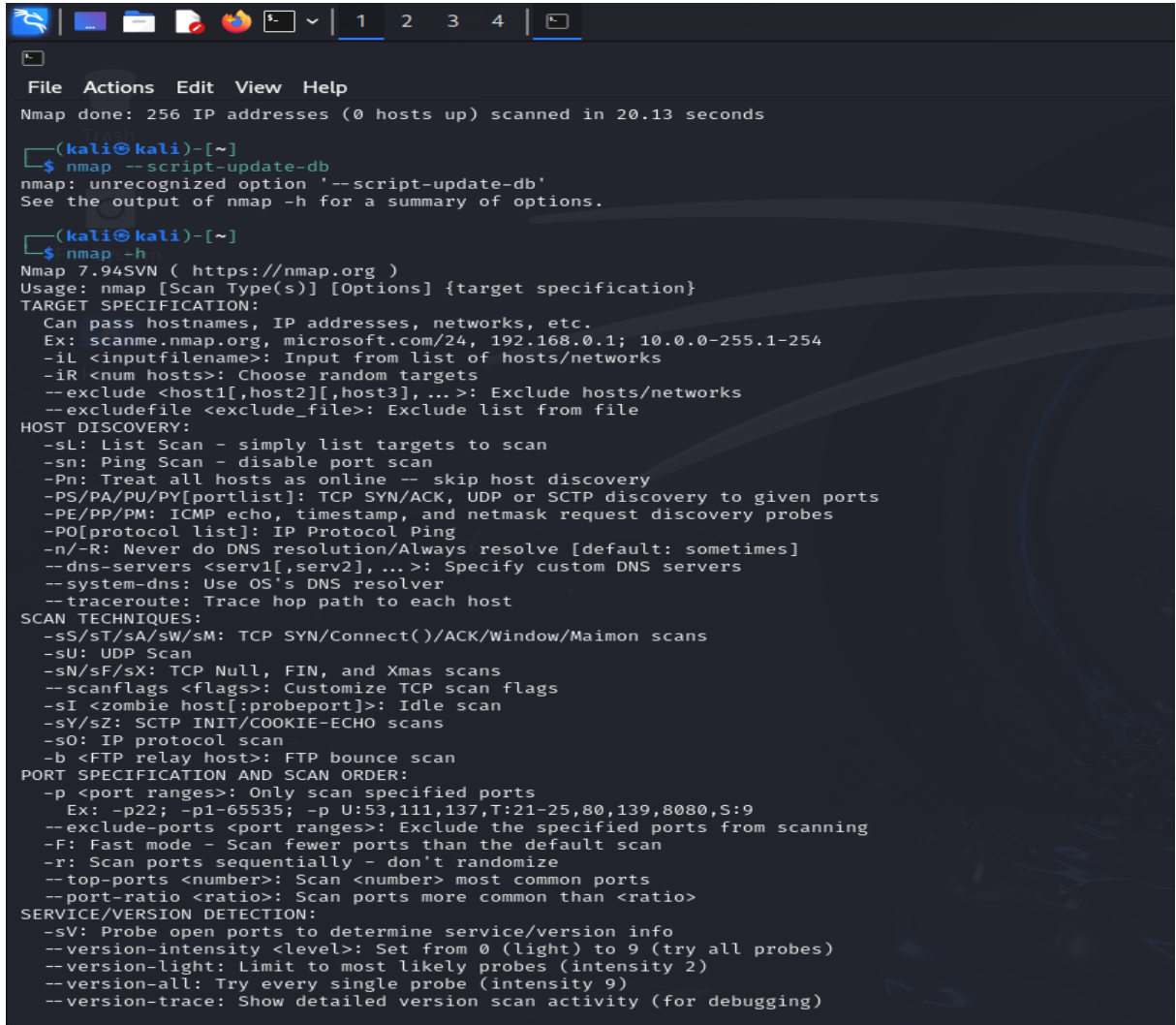
- execute the listed script against target IP address

```
(kali㉿kali)-[~]
$ nmap --script= test script 172.16.1.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:24 EST
Failed to resolve "test".
Failed to resolve "script".
Nmap done: 256 IP addresses (0 hosts up) scanned in 20.13 seconds

(kali㉿kali)-[~]
$
```

27. nmap --script-update-db :

- adding new scripts

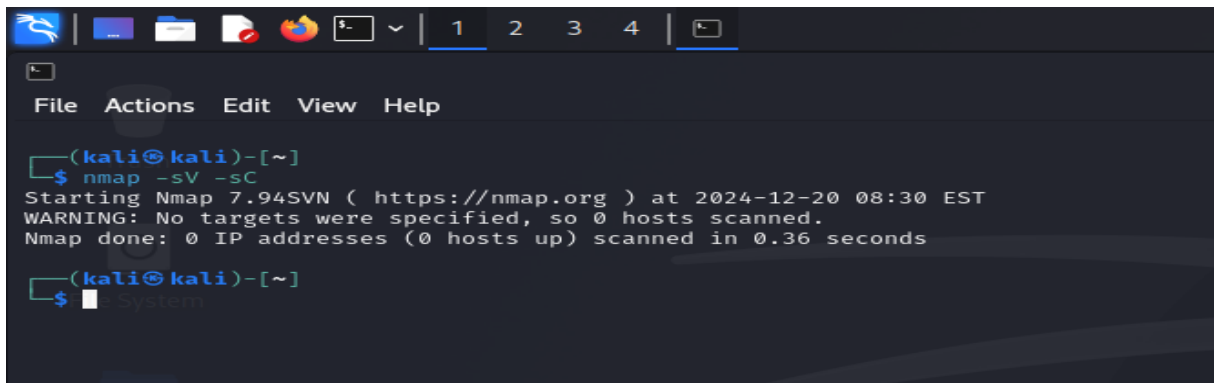


```
(kali㉿kali)-[~]
$ nmap --script-update-db
nmap: unrecognized option '--script-update-db'
See the output of nmap -h for a summary of options.

(kali㉿kali)-[~]
$ nmap -h
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PV/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
```

28. nmap -sV -sC :

- use of safe default scripts for scan



```
(kali㉿kali)-[~]
$ nmap -sV -sC
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:30 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.36 seconds

(kali㉿kali)-[~]
$
```

Date: 20/12/24

29. nmap -sV :

- try to find the version of the service running on port

```
(kali@kali)-[~]
$ nmap 172.16.1.1 -sV
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:34 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.11 seconds

(kali@kali)-[~]
$
```

30. nmap -sV --version-intensity :

- intensity level range 0 to 9

```
(kali@kali)-[~]
$ nmap 172.16.1.1 -sV --version-intensity 6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:38 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.13 seconds

(kali@kali)-[~]
$
```

31. nmap -sV --version -all :

- set intensity level to 9

```
(kali@kali)-[~]
$ nmap 172.16.1.1 -sV --version -all
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.2.2 libssh2-1.11.0 libz-1.3.1 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali@kali)-[~]
$
```

Date: 20/12/24

32. nmap -sV --version -light :

- Enable light mode

```
(kali㉿kali)-[~]
└─$ nmap 172.16.1.1 -sV --version -light
Nmap version 7.94SVN ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.6 openssl-3.2.2 libssh2-1.11.0 libz-1.3.1 libpcr2-10.42 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select

(kali㉿kali)-[~]
└─$
```

33. nmap -A :

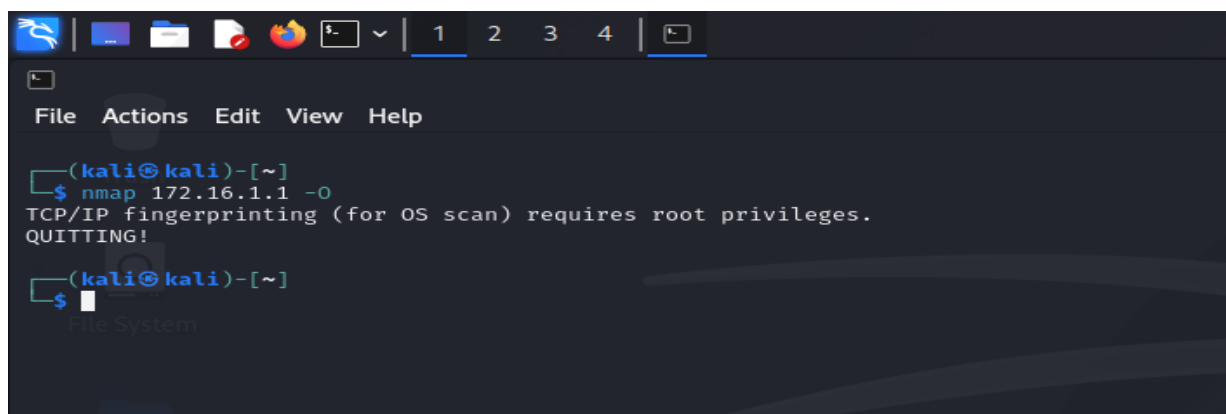
- enable OS detection, version detection, script scanning and traceroute

```
(kali㉿kali)-[~]
└─$ nmap 172.16.1.1 -A
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:48 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.18 seconds

(kali㉿kali)-[~]
└─$
```

34. nmap -O :

- remote OS detection



```
(kali㉿kali)-[~]
└─$ nmap 172.16.1.1 -O
TCP/IP fingerprinting (for OS scan) requires root privileges.
QUITTING!

(kali㉿kali)-[~]
└─$
```

Date: 20/12/24

35. nmap -oN scan.txt :

- default/ normal output

```
(kali㉿kali)-[~]
$ nmap -oN scan.txt 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:53 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.03 seconds

(kali㉿kali)-[~]
$ █
```

36. nmap -oX scanr.xml :

- XML

```
(kali㉿kali)-[~]
$ nmap -oX scanr.xml 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:55 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.03 seconds

(kali㉿kali)-[~]
$ █
```

37. nmap -oA:

- all format

```
(kali㉿kali)-[~]
$ nmap -oA 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 08:58 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.03 seconds

(kali㉿kali)-[~]
$ █
```

Date: 20/12/24

38. nmap -mtu [MTU] :

- specify MTU

```
(kali㉿kali)-[~]
$ nmap -mtu [MTU] [172.16.1.1]
Data payload MTU must be >0 and multiple of 8
QUITTING!

(kali㉿kali)-[~]
$ █
```

39. nmap -6 :

- scan IPV6 targets

```
(kali㉿kali)-[~]
$ nmap -6
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 10:32 EST
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.02 seconds

(kali㉿kali)-[~]
$ █
```

40. nmap 172.16.1.1 :

- single IP scan

```
(kali㉿kali)-[~]
$ nmap 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 10:37 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.05 seconds

(kali㉿kali)-[~]
$ █
```

Date: 20/12/24

41. nmap 172.16.1.1 172.16.100.1 :

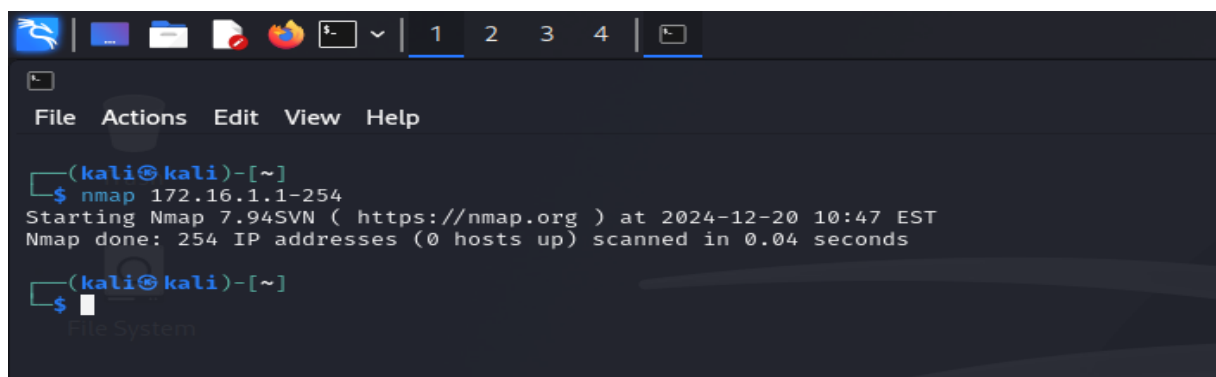
- scan specific IPs

```
(kali㉿kali)-[~]
$ nmap 172.16.1.1 172.16.100.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 10:44 EST
Nmap done: 2 IP addresses (0 hosts up) scanned in 0.02 seconds

(kali㉿kali)-[~]
$
```

42. nmap 172.16.1.1-254 :

- scan range of IP



```
(kali㉿kali)-[~]
$ nmap 172.16.1.1-254
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-20 10:47 EST
Nmap done: 254 IP addresses (0 hosts up) scanned in 0.04 seconds

(kali㉿kali)-[~]
$
```

43. nmap -sP :

- ping scan only

```
(kali㉿kali)-[~]
$ nmap -sP 172.16.1.1
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-21 00:50 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.04 seconds

(kali㉿kali)-[~]
$
```




Date: 20/12/24

44. nmap -PE :

- ICMP echo ping

```
(kali㉿kali)-[~]
$ nmap -PE 172.16.1.1
Warning: You are not root -- using TCP pingscan rather than ICMP
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-12-21 00:53 EST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 0.02 seconds

(kali㉿kali)-[~]
$ █
Home
```